

AI-ASSISTED CLINICAL CODE LIST RECOMMENDATION SYSTEM

Technical Case Study

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1. Executive Summary

This project explores the design and development of an AI-assisted system to support **clinical code list creation** in healthcare research workflows.

Developed as part of an employer-linked applied project with the National Institute for Health and Care Excellence (NICE), the solution addresses the challenge of identifying clinically relevant codes across large, fragmented terminology systems.

The final system implements a **multi-stage hybrid retrieval architecture**, combining semantic similarity search, ontology hierarchy traversal, benchmark validation, and LLM-based justification to produce **explainable, confidence-ranked clinical code recommendations**.

A key outcome of the project was a design pivot from a generative RAG-based approach to a **deterministic retrieval system**, prioritising reproducibility, auditability, and evaluation reliability in a regulated healthcare context.

2. Problem Context

Clinical research workflows often require defining patient cohorts using standardised clinical code lists (e.g., SNOMED CT).

However, this process is:

- manual and time-intensive
- dependent on specialist knowledge
- prone to inconsistency across studies
- difficult to reproduce and validate
- challenging when dealing with complex or multi-condition cohorts

Additionally, terminology systems may contain **hundreds of thousands of concepts**, making exhaustive search impractical.

The goal of this project was to design a system that:

- improves retrieval efficiency
- increases coverage of relevant codes (recall)
- maintains accuracy (precision)
- provides explainable recommendations
- supports structured clinical review

3. Initial Architecture: RAG-Based Approach

The project initially explored a **Retrieval-Augmented Generation (RAG)** architecture, where a Large Language Model (LLM) would:

1. interpret research queries
2. retrieve relevant information
3. generate candidate code lists

While flexible, this approach revealed a critical limitation:

Non-deterministic outputs

Repeated runs of the same query produced different results, making it difficult to:

- evaluate using precision/recall
- compare model performance
- ensure reproducibility
- support audit requirements

Given the importance of traceability in healthcare workflows, this approach was not suitable as the primary system design.

4. Design Pivot: Deterministic Retrieval

The system was redesigned to prioritise:

- deterministic outputs
- traceable evidence paths
- modular validation
- explainable decision logic

Rather than generating codes directly, the system focuses on:

retrieving, validating, and ranking candidate codes using multiple independent evidence sources

This shift significantly improved:

- evaluation reliability
- reproducibility
- auditability
- interpretability

5. Final System Architecture

The final architecture is a **multi-stage hybrid retrieval pipeline**:

Step 1 — Query Processing

- Free-text research queries are converted into structured keywords
- Supports flexible user input while enabling structured retrieval

Step 2 — Codelist Discovery

Candidate code lists are identified using:

- TF-IDF similarity
- SapBERT embeddings
- MiniLM embeddings

Primary source:

- OpenCodelists

This step prioritises curated groupings of clinically relevant codes.

Step 3 — Semantic Retrieval

- SapBERT embeddings are used to represent clinical concepts
- Vector similarity search is performed using FAISS

This enables identification of semantically similar clinical concepts beyond keyword matching.

Step 4 — Hierarchical Expansion

- SNOMED CT ontology structure is used
- Parent-child relationships are traversed using Lowest Common Ancestor (LCA) logic

This captures structurally related concepts that may not be directly retrieved via semantic similarity.

Step 5 — Validation & Confidence Scoring

Retrieved codes are validated against:

- NHS reference sets
- QOF clinical rules

A confidence framework is applied:

- **High Confidence** → supported by multiple evidence paths
- **Medium Confidence** → partial agreement
- **Low Confidence** → single-source evidence

This provides a structured prioritisation for clinical review.

Step 6 — LLM-Based Justification Layer

A justification layer was implemented using Google Gemini 2.5 Flash-Lite.

Two prompting strategies were explored:

- deterministic rule-based prompting
- zero-shot prompting

The LLM provides:

- reasoning for inclusion/exclusion
- contextual interpretation of clinical relevance
- explanation aligned with confidence classification

This separate:

- **evidence generation (retrieval)**
- **reasoning and explanation (LLM layer)**

improving transparency.

Step 7 — Output Design

The final output is delivered as a structured Excel file containing:

- full list of recommended codes
- evidence source columns (per retrieval stage)
- justification and reasoning
- confidence classification

Tabs include:

- full list
- high confidence
- medium confidence
- low confidence

This format supports efficient review and decision-making.

6. Data Sources & Knowledge Base

The system integrates multiple publicly available datasets:

- SNOMED CT (primary terminology system)
- OpenCodelists
- NHS reference sets
- QOF rules

A **master SNOMED dictionary** was constructed and enriched with:

- synonym mappings
- hierarchical relationships ("IS-A")
- ontology traversal structures

7. My Contribution

My primary contributions focused on system design, data integration, and validation:

Project Coordination

- coordinated cross-functional workstreams across data, modelling, and evaluation
- supported architecture decisions and development alignment

Knowledge Integration

- designed and built the SNOMED CT master dictionary
- integrated synonym expansion and hierarchy structures
- prepared data for semantic and hierarchical retrieval workflows

Validation Framework

- designed confidence scoring logic
- implemented validation using QOF rules and NHS reference sets
- contributed to explainable evidence pathways for clinical review

8. Key Challenges

Non-deterministic outputs

Resolved by moving away from generative RAG to deterministic retrieval.

Granularity mismatch

Code-level retrieval produced noisy results → shifted to codelist-first approach.

Data consistency

Validation gaps highlighted need for strict dictionary checks.

Cross-team alignment

Parallel development streams required structured coordination.

9. Limitations

- recall varies across conditions depending on synonym coverage
- hierarchy traversal depends on ontology completeness
- comorbidity handling not fully integrated
- LLM reasoning quality depends on prompt design
- system not deployed in production environment

10. Future Extensions

- integration of ICD-10 and OPCS-4
- persistent vector database (e.g., Chroma)
- enhanced synonym expansion
- improved multi-condition cohort modelling
- deployment as an interactive research support tool

11. Key Learnings

This project reinforced that:

- high-quality AI systems require **structured evaluation, not just modelling**
- retrieval systems benefit from **multiple complementary evidence sources**
- explainability is critical in regulated domains
- deterministic design is often essential for real-world adoption
- combining retrieval + validation + reasoning leads to stronger decision support systems

A key takeaway is that:

effective AI systems in healthcare must balance performance with transparency, reproducibility, and trust.